

Super Sleuths

(Scheduled Event - 30 Minutes)



Description: Teams will receive a crime scenario, a collection of evidence samples, and a list of possible suspects. Participants will perform laboratory tests, analyze physical evidence, and apply scientific reasoning, to identify the most likely suspect(s) and solve the crime.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Safety Equipment - Teams Must Bring:

- **Safety glasses labeled ANSI Z87+ (impact rated)**
- **All competitors must wear their eye protection during any competitor's test phase of the competition.**
- **If a team does not have the required eye protection, they will be given the opportunity to obtain it, time allowing, but will not receive extra time.**
- **If a team is unable to obtain eye protection, the team will not compete and will receive a no-show score.**

Indirect Vent Goggles Required



Eyeglasses are not safety rated!

Teams must bring:

- Pen/Pencil

Teams may not bring:

- Notes or guides of any kind.

Safety Requirements:

- Participants are expected to conduct themselves in a safe and responsible manner at all times.

General Safety Rules

- Follow all instructions provided by the Event Supervisor.
- Do not taste, smell directly, or intentionally ingest any substance provided during the event.
- Do not place any material near the eyes, nose, or mouth.
- Handle all evidence samples carefully and only as directed.
- Report spills, broken equipment, or unsafe conditions immediately to the Event Supervisor.
- Do not remove any materials from the competition area.
- Horseplay, misuse of materials, or unsafe behavior is prohibited.

Safety Penalties

- First minor safety violation: Warning.
- Additional minor safety violations: Up to 10% deduction from the final score per violation.

- Major safety violations (including intentional misuse of materials, tasting substances, or behavior that endangers participants or equipment): Tiered below teams with no major safety violations.
- Repeated or severe unsafe conduct: Disqualification at the discretion of the Event Supervisor.

Materials Provided at Event:

Each team will receive identical evidence samples and testing materials. Participants will be provided with:

- | | |
|-------------------|------------------------------|
| • Cups | • pH paper |
| • Spoons | • Magnifying glass |
| • Stirrers | • Crime scenario |
| • Magnet | • Evidence samples |
| • Deionized water | • Suspect information sheets |

The Competition:

1. Qualitative Analysis Station

Teams will identify unknown evidence by performing appropriate tests and observations, including solubility, magnetic properties, acidity (pH), color, texture, other observable characteristics.

- **Powders:** Cornstarch, Baking soda, Gelatin, Yeast, Sand, Ascorbic acid
- **Liquids:** Lemon juice, Hydrogen peroxide, Water, Soap solution
- **Metals:** Iron, Copper, Aluminum

2. Polymer Testing and Natural/Man-Made Substances

Teams will identify and analyze evidence from a variety of natural and manufactured materials.

- **Hair:** Difference between human, dog, and cat hair (microscopic pictures or descriptions will be provided)
- **Fibers:** Difference between animal, vegetable, and synthetic (microscopic pictures, source of fibers, or descriptions will be provided)
- **Recyclable Plastics:** HDPE, PETE, PMMA, PS

3. Other Evidence Analysis

Teams will identify and analyze additional forms of forensic evidence and answer related knowledge questions.

- **Fingerprints:** Identify different patterns on fingerprint evidence such as the difference between whorls, loops, and arches.
- **Shoe/Tire Prints:** Compare prints and make conclusions, including direction and speed.
- **DNA:** Compare DNA chromatograms from materials found on the scene to those of the suspects.

4. Crime Scene Analysis

Teams will use all evidence collected during the event to solve the crime. Participants will prepare a written analysis that:

- Identifies the suspect(s) most strongly supported by the evidence.
- Explains which evidence supports the conclusion.
- Describes how the evidence connects the suspect(s) to the crime.
- Explains why the remaining suspects were eliminated.
- Demonstrates logical and scientific reasoning based on the evidence provided.

Scoring:

- Accuracy of evidence identification
- Interpretation of forensic evidence
- Knowledge-based questions
- Crime scene analysis and conclusions
- Quality of scientific reasoning

Tiebreakers:

- Highest crime scene analysis score.
- Highest evidence identification score.
- Highest DNA analysis score.
- Event Supervisor-selected tiebreaker question(s).

Resources:

- [Crime Busters](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles